AI Hospital Appointment Prioritization

Project: AI Hospital Appointment Prioritize

Team Members:

- Karim Abo Zid: 60104642
- Mohsen Ali: 60305864
- Yousef Natsheh: 60307770
- Abdul Karrim Attar: 60100809
- Ameer Beitelmal: 60311435

Introduction

Patients and hospitals in Qatar face many challenges with the current healthcare system some of these challenges are long waiting time for patient appointments, difficulty in finding available time slots that fit them into the system, high rates of missing appointments, inefficient and hard way in manual scheduling for the receptionist staff. So, to solve this problem we came with the AI Hospital Appointment prioritize system.

AI Hospital Appointment prioritize is an AI platform that uses AI to intelligently prioritize and schedule hospital appointments in real time. It will do that by using predictive analytics and smart classification, The system will solve many problems that face the patients and the staff like reducing the waiting times, decreasing the missed appointments, optimizing the resources usages, and will ensure that the patients have appointments in efficient way. And project will directly support Qatar’s National health strategy 2024-2030.

Selected Strategic Goal from NHS-3

Our project will support two out of the three official NHS-3 Priority areas in a direct way

- First, priority that we will support is Excellence in service delivery and patient experience.
- Second, priority that we will support is the health system efficiency and resilience.

Project scope

In scope	Description
Patient booking portal	It will be a web based with a responsive interface that allows the patients to search, view, cancel and book their appointments online without the need to come or call the hospital.
AI prioritization system for booking and scheduling appointments	It is an intelligent system that analyses patient information and there needs while also providing for the urgent cases and earlier slots.
Prediction for missing appointments	It will be a machine learning system that analyses the patient historical

	Data to predict the chance of a patient to miss their appointment, this allows the system to take a proactive measure to reduce the damage.
Staff dashboard	It's for the hospital administrative staff, receptionist and the mangers to help them monitor, manage, and analyze the appointment data.
Basic integration with the existing hospital systems	Connect with the current information in the hospital to take essential data without the need to replace whole system
Automated SMS reminders	It is an automated SMS notification system that will send a reminder to patients before there coming appointments also to confirm their ability to come

Out of scope	Why it's out of the scope
Payment and billing processing	It's out of the scope because the payment systems need very good security infrastructure also it will need a lot of things to do, and the NHS-3 project do not focus on that part.
Wearable integration	Because there will be no point to integrate wearable devices and it will only make it more complex.
Equipment improvement	Because it's a project that focuses on the software not the hardware or facilities and it's the hospital responsible.

Project objectives

- Reduce patient appointment waiting time by 50% at least reduction in the days to wait until their appointment.

- Decrease no show rates by 40%: reduction on the missed appointment through the reminders and overbooking.

- Improve administrative efficiency: by reducing the time spent on scheduling appointments by at least 60%.

- Increase patient satisfaction: the target is to get positive feedback by at least 85% of the patients.

- Optimize doctor utilization: by lowering the unused slots by at least 35%.

Project constraints:

- First thing is that it must work with existing hospital infrastructure.
- Also, it must be ready for hospital expansion in the future if there was any.
- Finally, the AI prediction system must achieve at least 80% accuracy with the prediction.

Key users

Key users	Description	Key needs
Patients	Are the people that want to take appointments.	Easy booking, reminders, not far appointments.
Receptionists	Are the staff that mange the appointments for the patients.	Efficient booking tools, no show management.
Doctors	Are the people that provide healthcare.	Good schedules, reduces the empty spaces in the schedules that dose not have appointments, urgent slot management.

Hospital Administrators	Are the department managers.	Analytics on no show, patient wait time, and system performance metrics.

Key functionalities

- **Predictive no show risk analysis:** the system will use a machine learning system to analyze the patient historical data to predict the chance of the patient missing their appointment, it will do overbooking and send reminder messages for the patients with high probability of missing their appointments to have an efficiency use of the resources.

- **Automated reminders:** based on the probability of no show the system will automatically send reminder messages for example one or two days and also with the message it will have a link so if the patient can't come, he can cancel or reschedule the appointment

- **Staff dashboard**: it is a dashboard for the staff that will let them spend less time on scheduling appointments, also it transforms the data in the system into a readable way for better management

- **Appointment booking for urgent cases:** it is a booking portal that combines the availability of the slots with AI that evaluates the medical need of the patient and automatically prioritize urgent cases to take earlier appointments

AI powered features

- **First AI feature: intelligent appointment slot recommendation:** When the patient request appointment the AI will recommend the time slots based on: Patient preferred time which will be learned from the history, the urgency of the condition, doctor availability, the history of the no show, minimizing patient waiting time.

- **Second AI feature is: no show prediction**: it’s an AI model that will analyze the patient historical data like the past appointments that he came for, also the appointments that he didn’t come for to predict the possibility of a patient missing their scheduled appointment. So, for the patients with no high show, it will have extra things to do like double confirmation from the patient, and overbooking.

Requirements Document

Functional Requirements

Functional requirements describe what the system must do.

ID	Functional Requirements	Description	Priority
P1	Patient Registration	Allow patients to create an account in the system	High
P2	Patient Login	Allow patients to login to the system	High
P3	View Appointments	Allow patients to view appointments for chosen doctors and dates	High
P4	Book Appointments	Allow patients to book appointments by choosing available time slots	High
P5	Modify Appointment	Allow patients to change appointments time or selected doctor if available	Medium
P6	Cancel Appointment	Allow patients to cancel schedule appointments	Medium
P7	Submit Information	Allow patients to submit symptoms and health information to system	High
S1	AI Patient Analysis	Analyze patient data using an AI prioritization model	High
S2	Urgency Score Calculation	Calculate an urgency score for each appointment booked	High
S3	Appointment Prioritization	Prioritize appointments based on urgency score	High
S4	Time Slot Recommendation	Recommend suitable time slots based on urgency and availability	High
S5	No-Show Prediction	Predict patient missing their appointment	High
S6	Appointment Reminder	Send appointment reminders to patients	High

ID	Functional Requirements	Description	Priority
S7	Waitlist	Patients will be added to waitlist if appointment is fully booked	High
D1	Doctor Dashboard	Allow doctors to view prioritized patient appointments	High
D2	View Patient History	Allow doctors to view patient appointments history	High
R1	Receptionists Dashboard	Allow Receptionists to view prioritized patient appointments	High
R2	Manage Appointments	Allow Receptionists to manage patient appointment schedules	High
R3	Register Patient	Allow Receptionists to register patient	Medium
A1	Admin Dashboard	Allow Admins to view no-show rates, wait times, and utilization metrics	Medium

Non-Functional Requirements

Non-functional requirements describe how the system performs.

ID	Non-Functional Requirements	Priority
1	The system shall respond quickly to user requests	High
2	The system shall securely store patient data	High
3	The system must be available 24/7	High
4	The system shall protect patient data privacy	High
5	AI predictions should be generated in 5 seconds	High
6	System should support large number of users at the peak times	High

Aa ID	Non-Functional Requirements	Priority
7	The system interface must be clear and easy to use	Meduim
8	System must be maintainable and easy to update	Meduim
9	System should be accessible on mobile	Meduim
10	Booking process should be completed in less than 4 steps	Meduim
11	Interface shall support Arabic and English	Meduim

Work breakdown structure (WBS)

Task ID	Task Name	Type	Assigned To	Duration	Depends On	Status
1	Planning Phase	Planning	Yousef Tariq Natsheh MohsenAli	4	-	Done
1.1	Project Plan	Planning	Yousef Tariq Natsheh	1	-	Done
1.2	Requirments	Planning	MohsenAli Yousef Tariq Natsheh	1	1.1	Done
1.3	Notion workspace	Planning	MohsenAli	1	-	Done
1.4	WBS and Timeline	Planning	MohsenAli	1	1.1	Done
2	Analysis Phase	Analysis	Ameer Beitelmal Abdalkareem Attar MHDKarim Abozid	4	1	Not started
2.1	Use case model	Analysis		2	1.2	Not started
2.2	Wireframes	Analysis		1	2.1	Not started
2.3	Class Diagram	Analysis		1	2.1	Not started

Task ID	Task Name	Type	Assigned To	Duration	Depends On	Status
3	Design Phase	Design	Ameer Beitelmal Abdalkareem Attar MHDKarim Abozid	7	2	Not started
3.1	System Architecture	Design		2	2.3	Not started
3.2	UML Design Class Diagram	Design		2	3.1	Not started
3.3	REST API Design	Design		1	3.1	Not started
3.4	Sequence Diagrams	Design		1	3.2	Not started
3.5	Administration Evaluation	Design		1	3.3	Not started
4	AI functionalities	AI	MohsenAli Yousef Tariq Natsheh	8	3	Not started
4.1	AI Feature 1	AI		2	3.4	Not started
4.2	AI Feature 2	AI		2	3.4	Not started
4.3	AI Prompts	AI		1	4.1	Not started

Task ID	Task Name	Type	Assigned To	Duration	Depends On	Status
4.4	Inputs / Outputs	AI		1	4.1	Not started
4.5	Documentation	AI		1	4.2	Not started
4.6	Project presentation	AI		1	4.5	Not started

Project Timeline

Week	Phase	Tasks	Date
Week 11	Planning	Project plan + requirements	@March 15, 2026
Week 13	Analysis	Use case + wireframes + class diagram	@March 29, 2026
Week 14	Design	Architecture + REST API + sequence diagrams	@April 5, 2026
Week 15	Ai	AI prompts + final presentation	@April 12, 2026